



Federal Office
for Information Security

Technical Guideline TR-03105

Test Plan for Official Electronic ID Documents with Secure Contactless Integrated Circuit

Part 2

Version 3.0.1 RC1



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Table .1 Document history

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1 Introduction

Passports are a key element of travels between countries. Although the individual document styles differ – depending on the issuing country – it has to be ensured that each passport's construction follows several general rules, in order to guarantee interoperability. These general rules are defined in [ICAO 9303], published by ICAO.

Traditionally, passports only contained printed data (partially, in machine-readable form) but with the evolving technology of contactless smart card systems it was decided (at the beginning of the 21st century) to allow integration of a contactless chip (together with an appropriate antenna) into the passport document. Consequently, this additional element is also defined in (an extension to) [ICAO 9303].

The underlying technology of a so-called “ePassport” is defined in the ISO/IEC 14443 standard series (parts 1 ([ISO/IEC 14443-1:2016]) to 4 ([ISO/IEC 14443-4:2016])) covering aspects starting from physical characteristics up to transmission protocol definitions), extended by an application specification defined in [ICAO 9303].

In order to ensure that a given (e)Passport matches its specifications, it is of crucial importance to perform standardized tests. In addition to “traditional” tests of the document itself (e.g. durability tests) the inclusion of smart card technology makes it necessary to test these components, too. For this reason, the specification “TR-03105” was issued by the BSI (initially, in 2005).

BSI TR-03105 is a series of documents which is split up in general requirements (introduction and Part 1), card-testing (Parts 2 and 3.x) and reader-testing (Parts 4 and 5.x) aspects. Both card-testing and reader-testing topics consist of lower-layer (Parts 2 and 4) and application-layer (Parts 3.x and 5.x) parts.

Generally, the TR-03105 documents form the basis of certification testing for German ePassports (and, as introduced later, German eID cards) documents and readers. But they can also be applied to foreign ePassports and eID products (as long as it is guaranteed that they are constructed in a standardized way) – thus giving the TR-03105 an importance beyond the German market.

Within this specification, BSI TR-03105 Part 2, the focus is on lower-layer (meaning ISO/IEC 14443-1 .. -4) document testing. Both earlier editions of this specification were based on [ISO/IEC 10373-6:2001] (the first edition of the general test standard for ISO/IEC 14443-compliant products) but this third edition is based on the new ePassport testing standard [ISO/IEC 18745-2:2016].

The update of TR-03105 Part 2 was necessary in order to ensure that all current technology – ISO/IEC 14443 significantly changed and expanded during the past years – is still covered by this test specification. While former editions contained many unique test case definitions (which were not covered by [ISO/IEC 10373-6:2001] at that time), the new third edition is mainly referring to matching [ISO/IEC 18745-2:2016] test scenarios (with only few own definitions remaining) in order to ensure a high compatibility to this standard.

1.1 Terminology

The key words "MUST", "SHALL", "REQUIRED", "SHOULD", "RECOMMENDED", and "MAY" in this document are to be interpreted as described in [RFC 2119].

Key word	Description
MUST	This word, or the terms "REQUIRED" or "SHALL", mean that the definition is an absolute requirement of the specification
MUST NOT	This phrase, or the phrase "SHALL NOT", mean that the definition is an absolute prohibition of the specification.
SHOULD	This word, or the adjective "RECOMMENDED", mean that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
SHOULD NOT	This phrase, or the phrase "NOT RECOMMENDED" mean that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.

Key word	Description
MAY	This word, or the adjective "OPTIONAL", mean that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include a particular option MUST be prepared to interoperate with another implementation which does include the option, though perhaps with reduced functionality. In the same vein an implementation which does include a particular option MUST be prepared to interoperate with another implementation which does not include the option (except, of course, for the feature the option provides.)

Table 1.1 Terminology

2 General Test Requirements

2.1 Validity of Referenced Documents

In order to prevent potential conflicts caused by updates of any of the referenced documents (see Annex), no modifications or extensions of these documents occurring after October 01, 2016 are taken into account. This especially holds for (yet unpublished) amendments to specifications which would have automatically become applicable right after publication otherwise.

2.2 *PICC* Class Limitation

Throughout this whole document, it is always assumed that a *DUT* complies to *PICC* class 1 definition – it is not considered necessary to additionally cover further *PICC* classes in this context¹. Thus all test scenarios (especially in Layer 2) **MUST** be performed such that only *PICC* class 1 test setups and limit values are applied.

2.3 Test Setup and Equipment

In general, the definitions given in [ISO/IEC 18745-2:2016], Chapters 5.1 (“General test conditions”) and 5.1.1 (“Test setup”) **SHALL** be used without modifications.

If a certain test case requires a different setup, it **SHALL** be specified in the context of the individual test case definition

2.4 Nominal Values

Unless stated otherwise, the values defined in [ISO/IEC 18745-2:2016], Chapter 5.1.2 (“Values unless otherwise specified”) and Chapter 7.1, Table 6 (“Values for bit rates higher than $f_c/16$ ”) **SHALL** be used without modifications.

2.5 Report

The test report **SHALL** include the number of passed tests versus the total number of tests. A description of each test, the information if the test was **PASS** or **FAIL**, the number of different samples and the date of the tests **MUST** be included.

In addition to this general claim, further requirements may be given in the definitions of the individual test scenarios.

2.6 Test Sequence

As the test sequences highly depend on the test scenarios, they are part of the individual test case definitions in Chapters 2 to 6.

General test commands and command sequences (used for all Layer 4 and several Layer 2 and Layer 3 test scenarios) are defined in Annex B.

¹ As ePassports and eID cards typically are equipped with “large” antennas the limitation to only support *PICC* class 1 is considered reasonable in the context of this document. In case that a *DUT* should explicitly claim compliance to any other *PICC* class, it still shall be tested according to *PICC* class 1 rules (without considering any additional test parameters which might be applicable to the claimed *PICC* class).

3 Layer 1 Tests

As defined in [ISO/IEC 18745-2:2016], Chapter 5.1, all tests within this group SHALL be performed with 3 (identical) samples. All 3 test performances MUST be successfully completed for a test scenario to count as PASS.

3.1 "Class 1" Verification (Conditional)

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.2.1. It SHALL be performed as described there, without modifications.

3.2 Static Electricity

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.2.2. It SHALL be performed as described there, without modifications.

3.3 Alternating Magnetic Field

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.2.3. It SHALL be performed as described there, without modifications.

4 Layer 2 Tests

As defined in [ISO/IEC 18745-2:2016], Chapter 5.1, all tests within this group SHALL be performed with 3 (identical) samples. All 3 test performances MUST be successfully completed for a test scenario to count as PASS.

4.1 Load Modulation Amplitude

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.3.1 (“eMRTD transmission”). It SHALL be performed as described there, without modifications.

4.2 Operating Field Strength

Test-ID	ISO/IEC_18745_Layer2_2
Reference	[ISO/IEC 18745-2:2016], Chapter 5.3.2 (“Operating field strength”) and Chapter 7.2.1.
Version	
Executions conditions	This test case SHALL be performed
Purpose	The purpose of this test is to check if the eMRTD operates its application within the operating field strength range defined in [ISO/IEC 14443-2:2016].
Preconditions	
Test scenario	This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.3.2 (“eMRTD reader field strength”) and Chapter 7.2.1. Depending on the bit rate capabilities of the DUT, • [no bit rates higher than $f_c/16$ are supported] only the scenario defined in Chapter 5.3.2 SHALL be performed • [otherwise] both variants of the test case SHALL be performed In any case, no modifications to the [ISO/IEC 18745-2:2016] scenario(s) SHALL be applied.
Expected results	The PICC was able to operate its application under all applicable conditions.
Remarks	
Verdict	The test result MUST be evaluated as PASS or FAIL.

Table 4.1 Test case ISO/IEC_18745_Layer2_2

4.3 SCIC Reception

Test-ID	ISO/IEC_18745_Layer2_3
Reference	[ISO/IEC 18745-2:2016], Chapter 5.3.3 (“eMRTD Reception”) and Chapter 7.2.2.
Version	
Executions conditions	This test case SHALL be performed
Purpose	The purpose of this test is to verify the ability of the PICC to receive the PCD commands.
Preconditions	
Testscenario	This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.3.3 and Chapter 7.2.2. Depending on the bit rate capabilities of the DUT, • [no bit rates higher than $f_c/16$ are supported] only the scenario defined in Chapter 5.3.3 SHALL be performed • [otherwise] both variants of the test case SHALL be performed The following modification to the referenced specifications SHALL be applied: • Instead of only using TEST_COMMAND1 the test MUST be performed using TEST_COMMAND_SEQUENCE1 (refer to the “Operating Field Strength” test for further details on how to exactly use this command sequence).

Test-ID	ISO/IEC_18745_Layer2_3
Expected results	The PICC was able to receive the PCD commands in all applicable conditions.
Remarks	
Verdict	The test result MUST be evaluated as PASS or FAIL.

Table 4.2 Test case ISO/IEC_18745_Layer2_3

4.4 SCIC Resonance Frequency (Optional)

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.3.4. It SHALL be performed as described there, without modifications.

4.5 SCIC Maximum Loading Effect

Test-ID	ISO/IEC_18745_Layer2_5
Reference	[ISO/IEC 18745-2:2016], Chapter 5.3.5 ("eMRTD Maximum Loading effect")
Version	
Executions conditions	This test case SHALL be performed
Purpose	The purpose of this test is to measure the PICCs loading capability.
Preconditions	
Testscenario	This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.3.5. It SHALL be performed as described there, with the following modification: In addition to the specified test procedure (where $V_{load} = 6\text{ V}$ MUST be set, as defined in [ISO/IEC 10373-6:2016], Chapter 7.2.5.2), the test SHALL be performed a second time with $V_{load} = 4.5\text{ V}$ (in order to maximize interoperability with further specifications).
Expected results	The measured field strength is higher than H_{min} .
Remarks	
Verdict	The test result MUST be evaluated as PASS or FAIL.

Table 4.3 Test case ISO/IEC_18745_Layer2_5

4.6 Low Field Recover Capability

Test-ID	
Reference	
Version	
Executions conditions	This test case SHALL be performed at multiple field strengths, starting from 0.1 A/m and continue – in steps with at most 0.1 A/m width – until H_{min} is reached.
Purpose	This test SHALL verify the correct <i>DUT</i> behavior at field strengths below the minimum required operating field strength H_{min} . The purpose of the test is to verify whether the <i>DUT</i> is capable of recovering from already started and unexpectedly interrupted communication processes at low field strengths.
Preconditions	
Testscenario	In order to perform this test, test procedure and test conditions from the "Operating Field Strength" test (see Chapter 4.2) SHALL be used, with the following modifications: Independent from the actual bit rate capabilities of the <i>DUT</i>, the test SHALL only be performed at $f_c/128$. 5 seconds after completion of the test scenario, a functionality check – using the procedure from the "Operating Field Strength" test – SHALL be performed at H_{min}. The <i>DUT</i> MUST successfully complete this procedure in order to PASS the test.

Test-ID	
Expected results	During this test, the <i>DUT</i> MAY not operate correctly but it MUST be able to recover from any state and operate normally after applying a field strength, which is in line with the normal operating conditions. The <i>DUT</i> MUST successfully complete this procedure in order to PASS the test.
Remarks	
Verdict	The test report SHALL state whether or not the <i>DUT</i> was able to recover from exposure to low field strengths. The <i>DUT</i> MUST successfully complete this procedure in order to PASS the test.

Table 4.4 Test Case Low Field Recover Capability

4.7 SCIC EMD Level and Low EMD Time

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 7.2.3. It SHALL be performed as described there, without modifications.

5 Layer 3 Timing Tests

Unless further specified, all tests within this group SHALL be performed

- with 3 (identical) samples
- at RT
- using the following field strengths: H_{min} , 2.5 A/m, 3.5 A/m, 4.5 A/m, 5.5 A/m, 6.5 A/m, H_{min}

5.1 Frame Delay Time (Type A only)

The purpose of this test is to verify that the *DUT* respects the *FDT* requirements as defined in [ISO/IEC 14443-3:2016], Chapter 6.2.1.1. The test SHALL be performed for various combinations of initial card states and reader commands:

Condition	Initial State	Command
1	IDLE	REQA
2	IDLE	WUPA
3	READY(1)	AC (ends with '0')
4	Ready(1)	AC (ends with '1')
5	Ready(1)	SELECT
6	HALT	WUPA
7	Ready*(1)	AC (ends with '10')
8	Ready*(1)	AC (ends with '1')
9	Ready*(1)	SELECT

Table 5.1 FDT Test Conditions

The following procedure **MUST** be performed for all 9 conditions defined in ▶Table 5.1:

1. The *DUT* is brought to the test initial state (by using the method defined in [ISO/IEC 10373-6:2016], Chapter G.3.3.2.1)
2. The test apparatus (*PCD* simulator) sends the defined command
3. Measure the *FDT* (*PCD* to *PICC*) between sent command and received *DUT* response (if any)
4. Deactivate *RF* field and re-start the procedure (beginning from step 1) until 10 runs were performed

In order to PASS this test, the following criteria **MUST** be met:

- The *DUT* **MUST** send a formally correct response to all provided commands
- Each of the 90 individual measurements SHALL deliver *FDT* values within the specified limits

5.2 Detailed Type B Timing Measurements

The purpose of this test is to verify that the *DUT* is able to handle *PCD* frames with different timings (*SOF*, *EOF*, *EGT*) and that its responses use correct timing values (*SOF*, *EOF*, *EGT*, *TR0*, *TR1*) as well.

In order to perform this test, the procedure defined in [ISO/IEC 10373-6:2016], Chapter G.4.3 (“Scenario G.22: *PICC* Reception”) SHALL be used, with the following modifications:

- For each set of frame parameters, 10 consecutive measurements SHALL be performed
- The test SHALL be performed with 3 (identical) samples

All of the performed test runs **MUST** deliver a positive result for the test to be considered as PASS.

5.3 Timing before *PCD SOF (TR2)* (Type B only)

The purpose of this test is to verify that the *DUT* respects the *TR2* requirements as defined in [ISO/IEC 14443-3:2016], Chapter 7.1.7.

In preparation of the actual test performance, the following initial procedure SHALL be performed:

1. The *RF* power of the test apparatus (*PCD* simulator) is activated, bringing the *DUT* to IDLE state
2. The test apparatus sends *REQB* command
3. From the *ATQB* response of the *DUT*, the bits b3 and b2 of the “Protocol_Type” field are extracted in order to determine the minimum *TR2* coding

After setting the appropriate *TR2* value, the following procedure MUST be performed:

1. The *RF* power of the test apparatus (*PCD* simulator) is activated, bringing the *DUT* to IDLE state
2. The test apparatus sends *REQB* command
3. The *DUT* is expected to respond with a formally correct *ATQB*
4. The test apparatus sends *ATTRIB* with minimum delay (using the *TR2* value determined earlier)
5. Record the presence of the *DUT* response (if any)
6. Deactivate *RF* field and re-start the procedure (beginning from step 1) until 10 runs were performed

In order to PASS this test, the *DUT* always MUST provide a formally correct response to the *ATTRIB* command from step 4

6 Layer 3 and Layer 4 Protocol Tests

The test scenarios defined in [ISO/IEC 18745-2:2016], Chapter 5.4 (“Test of ISO/IEC 14443-3 and ISO/IEC 14443-4 parameters”) and Chapter 7.4 (“Additional tests of ISO/IEC 14443-3 and ISO/IEC 14443-4 parameters for *PICC*”) SHALL be performed as described there, without modifications.

Appendix A Implementation Conformance Statement

In order to ensure a proper test setup, an applicant **MUST** provide a completed *ICS* (Implementation Conformance Statement) document.

“Table 2” as defined in [ISO/IEC 18745-2:2016], Chapter 5.1.4 (“Applicant declaration”) **MAY** be used as a template but the final definition of actual design and content of the *ICS* form is the task of the test laboratory (in coordination with the BSI).

In any case, the *ICS* **SHALL** contain the following paragraph: “In case that Extended Access Control (*EAC*) is supported, the following files (in the formats as defined below) **MUST** please be provided:

- Document Verifier (*DV*) certificate in the format specified in [TR-03110]; this certificate **MUST** to be signed with the *CVCA* private key (which **MUST** correspond to the *CVCA* public key stored on the *DUT*'s chip)
- Inspection System (*IS*) certificate in the format specified in [TR-03110]; it **MUST** be signed with the *DV* private key (which **MUST** match the public key as contained in the *DV* certificate)
- *IS* private key in pkcs8 format (it **MUST** match the public key contained in the *IS* certificate)”

In addition to the specification of the technical characteristics of the *DUT*, the *ICS* document **SHALL** contain the following information:

- Applicant's company name and address
- Contact partner (including name, phone number, and email address)
- *DUT*'s product identifier, hardware version, and software version

It **MUST** be ensured that the completed *ICS* (at least the technical parts) is provided to the test lab prior to the test start.

Appendix B List of Test Command Sequences

B.1 General

This annex contains definitions of test commands and test command sequences on application level; these are used by multiple test scenarios defined in this document: All Layer 4 tests and several Layer 2 and Layer 3 tests.

B.2 Test Commands for *SCIC* without Access Control (Plain)

The definitions of test command sequence and test commands given in [ISO/IEC 18745-2:2016], Chapter 5.5.1 SHALL be used without modifications.

B.3 Test Commands for *SCIC* Supporting *BAC*

The definitions of test command sequence and test commands given in [ISO/IEC 18745-2:2016], Chapter 5.5.2 SHALL be used without modifications.

B.4 Test Commands for *SCIC* Supporting *PACE*

The definitions of test command sequence and test commands given in [ISO/IEC 18745-2:2016], Chapter 5.5.3 SHALL be used without modifications.

B.5 Test Commands for *SCIC* Supporting *EAC*

The definitions of test command sequence and test commands given in [ISO/IEC 18745-2:2016], Chapter 8.2.1 SHALL be used without modifications.

B.6 Test Commands for *SCIC* Supporting *PACE* and *EAC*

The definitions of test command sequence and test commands given in [ISO/IEC 18745-2:2016], Chapter 8.2.2 SHALL be used, with the following modification:

- If the *DUT* supports *PACE-CAM* (Chip Authentication Mapping), the following definition of TEST_COMMAND_SEQUENCE1 (instead of the one contained in ISO/IEC) MUST be used:

Step	Command	Protocol
1	READ BINARY (EF.CardAccess)	
2	MSE: Set AT	PACE-CAM
3	GENERAL AUTHENTICATE (Encrypted Nonce)	PACE-CAM
4	GENERAL AUTHENTICATE (MAP Nonce)	PACE-CAM
5	GENERAL AUTHENTICATE (Perform Mutual Key Agreement)	PACE-CAM
6	GENERAL AUTHENTICATE (Mutual Authentication)	PACE-CAM
7	<i>Optional</i> : READ BINARY (EF.CardSecurity)	PACE-CAM
8	SELECT (Passport Application)	
9	READ BINARY (EF.CVCA)	TA(v1)
10	MSE: Set DST (DV)	TA (v1)
11	PSO: Verify Certificate (DV)	TA (v1)
12	MSE: Set DST (IS)	TA (v1)
13	PSO: Verify Certificate (IS)	TA (v1)
14	MSE: Set AT	TA (v1)
15	GET CHALLENGE	TA (v1)
16	EXTERNAL AUTHENTICATE	TA (v1)

Step	Command	Protocol
17	READ BINARY (EF.DG3)	

Table B.1 TEST_COMMAND_SEQUENCE1 (with PACE/EAC, PACE-CAM in use)

The ISO/IEC definitions of TEST_COMMAND1, TEST_COMMAND2, and TEST_COMMAND3 SHALL be used without modification (independent from the usage of *PACE-CAM*).

B.7 Test Commands for SCIC supporting *PACE* and *EAC2*

In addition to the “regular” *EAC* version (currently in use by many ePassports, also used within [ISO/IEC 18745-2:2016]) exists a second variation (named “*EAC2*” within this document), defined by the BSI ([TR-03110])

EAC2 – which is, e.g., used by German nPA (national ID card) documents – significantly differs from the “normal” *EAC*, thus it is required to handle it separately as modified definitions of the test command sequence have to be introduced.

B.7.1 Test Command Sequence

Two variations of *EAC2* MUST be regarded: The “Chip Authentication” (*CA*) protocol MAY either be used in version 2 or version 3 – depending on that, different test command sequences MUST be defined.

In the following, both test command sequence options shall be introduced.

Step	Command	Protocol
1	READ BINARY (EF.CardAccess)	
2	MSE: Set AT	PACE
3	GENERAL AUTHENTICATE (Encrypted Nonce)	PACE
4	GENERAL AUTHENTICATE (MAP Nonce)	PACE
5	GENERAL AUTHENTICATE (Perform Mutual Key Agreement)	PACE
6	GENERAL AUTHENTICATE (Mutual Authentication)	PACE
7	MSE: Set DST (DV)	TA (v2)
8	PSO: Verify Certificate (DV)	TA (v2)
9	MSE: Set DST (IS)	TA (v2)
10	PSO: Verify Certificate (IS)	TA (v2)
11	MSE: Set AT	TA (v2)
12	GET CHALLENGE	TA (v2)
13	EXTERNAL AUTHENTICATE	TA (v2)
14	READ BINARY (EF.CardSecurity)	
15	MSE: Set AT	CA (v2)
16	GENERAL AUTHENTICATE (Chip Authentication)	CA (v2)
17	SELECT (Passport Application)	
18	READ BINARY (EF.DG2)	
19	OPTIONAL: READ BINARY (EF.DG3)	

Table B.2 TEST_COMMAND_SEQUENCE1 (with PACE/EAC, CA v2 in use)

Step	Command	Protocol
1	READ BINARY (EF.CardAccess)	
2	MSE: Set AT	PACE
3	GENERAL AUTHENTICATE (Encrypted Nonce)	PACE

Step	Command	Protocol
4	GENERAL AUTHENTICATE (MAP Nonce)	PACE
5	GENERAL AUTHENTICATE (Perform Mutual Key Agreement)	PACE
6	GENERAL AUTHENTICATE (Mutual Authentication)	PACE
7	MSE: Set DST (DV)	TA (v2)
8	PSO: Verify Certificate (DV)	TA (v2)
9	MSE: Set DST (IS)	TA (v2)
10	PSO: Verify Certificate (IS)	TA (v2)
11	MSE: Set AT	TA (v2)
12	GET CHALLENGE	TA (v2)
13	EXTERNAL AUTHENTICATE	TA (v2)
14	READ BINARY (EF.CardSecurity)	
15	MSE: Set AT	CA (v3)
16	GENERAL AUTHENTICATE (Perfrom Key Agreement)	CA (v3)
17	MSE: Set AT	CA (v3)
18	GENERAL AUTHENTICATE (Compute Pseudonymous Signature)	CA (v3)
19	SELECT (Passport Application)	
20	READ BINARY (EF.DG2)	
21	<i>OPTIONAL</i> : READ BINARY (EF.DG3)	

Table B.3 TEST_COMMAND_SEQUENCE1 (with PACE/EAC, CA v3 in use)

B.7.2 Test Commands

The definitions of TEST_COMMAND1, TEST_COMMAND2, and TEST_COMMAND3 are equal to those in the “SCIC supporting *PACE*” scenario (see Annex B.4 of this document or [ISO/IEC 18745-2:2016], Chapter 5.5.3).

In cases where TEST_COMMAND2 is used, it **MUST** be ensured that both *PACE* and *EAC2* **MUST** be successfully performed before the defined test command can be carried out.

Appendix C Functionality Check

C.1 General

This annex specifies two optional tests to verify the *SCIC*/ePassport functionality on electrical and application level.

C.2 Application Functionality Check (OPTIONAL)

This test scenario is defined in [ISO/IEC 18745-2:2016], Chapter 5.6.2. It SHALL be performed as described there, without modifications.

C.3 Electric Functionality Check (OPTIONAL)

In any case, the “Operating Field Strength” test (as defined in Chapter 4.2) SHALL be performed. If the “Resonance Frequency” test (as defined in Chapter 4.4) is applicable, it SHALL be additionally performed.

Keywords and Abbreviations

AC	Anticollision Command
ATQB	Answer to request, type B
ATTRIB	PICC selection command, Type B
BAC	Basic Access Control
CA	Chip Authentication
CAM	Chip Authentication Mapping
CVCA	Card Identifier
DUT	Device Under Test
DV	Document Verifier
EAC	Extended Access Control
EGT	Extra Guard Time
EMD	Electromagnetic disturbance
eMRTD	electric Machine Readable Travel Document
EOF	End Of Frame
fc	Carrier frequency (13.56 MHz)
FDT	Frame delay time
H_{min}	Minimum operating field strength
ICAO	International Civil Aviation Organization
ICS	Implementation Conformance Statement
IS	Inspection System
PACE	Password Authenticated Connection Establishment
PCD	Proximity Coupling Device
PICC	Proximity Integrated Circuit Card
REQA	Request command, type A
REQB	Request command, type B
RF	Radio frequency
RT	Room temperature
SCIC	Secure Contactless Integrated Circuit
SOF	Start Of Frame
TR0	Guard time between the end of a PCD transmission and the start of the SCIC subcarrier generation

TR1	Synchronization time between the start of the SCIC subcarrier generation and the start of the SCIC subcarrier modulation
TR2	Synchronization time between the start of the SCIC's EOF and the start of the PCD's next SOF
V_{load}	DC voltage measured at connector CON3 of the Reference <i>PICC</i>
WUPA	Wake-UP Command, Type A

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